

The StockMixer Model for Portfolio Construction

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Abstract

This document describes the StockMixer model for monthly portfolio construction, which follows the mixing architecture of Fan and Shen (2024). The model exchanges information first across characteristics and then across the cross section through explicit stock to market and market to stock steps, and it is fitted under a maximum Sharpe ratio regression objective. We develop the full model, present the two mixing operations, explain why the time mixing component of the original architecture is omitted on this panel, and report the realised performance.

1 Overview and motivation

The architecture borrows the mixing paradigm, which builds a network entirely from operations that exchange information along one axis of a data tensor at a time. In the original StockMixer the axes are the features, the stocks, and time, and a separate mixing block operates along each. Here we adopt the feature axis and the stock axis, which correspond respectively to exchanging information across the characteristics of a security and across the securities of a month, and we retain the maximum Sharpe ratio regression objective and the ranked characteristic representation. The purpose is to couple the cross section through a learned exchange between the securities and a summary market state, which is more expressive than a fixed pooled average.

2 The data pipeline and objective

The raw inputs are a panel of monthly characteristics and a list of candidate features, and the target is the excess return of a security over the following month. A characteristic is retained if its coverage is at least ninety percent over the pre test period, which makes the selection causal and algorithmic. For each month every retained characteristic is transformed to its within month percentile rank, missing entries are set to the median before centring so that the inputs lie in the interval $[-0.5, 0.5]$, and securities missing more than one third of the characteristics are dropped. The network is fitted under the loss $(1 - \sum_i w_i r_i)^2$, which drives the realised managed return towards unity in every month, on a rolling window of at most sixty months, with a fifty epoch cold start, ten epoch warm starts thereafter, three seeds, and seed averaging of the weights normalised to unit gross exposure.

3 Indicator mixing

The first mixing operation acts along the characteristic axis. For each security independently, a small perceptron mixes the components of the input vector, so that information is exchanged across the characteristics of that security. Writing x_i for the input vector of security i , the indicator mixing step is the residual update

$$x_i \leftarrow x_i + \text{mix}(\text{LN}(x_i)), \quad (1)$$

where LN is layer normalisation and mix is a two layer perceptron whose input and output dimensions both equal the number of characteristics. This step allows the network to form interactions among characteristics before any cross sectional information is introduced.

4 Stock mixing through a market state

The second mixing operation acts along the stock axis and is the mechanism that couples the cross section. Each security is first tokenised to an embedding $z_i \in \mathbb{R}^d$. A stock to market step pools the tokens into a single market state,

$$m = \frac{1}{n} \sum_{i=1}^n \text{to_market}(z_i), \quad (2)$$

and a market to stock step broadcasts that state back to every security and updates its token,

$$z_i \leftarrow z_i + \text{from_market}([z_i, m]). \quad (3)$$

The score is then read from a head applied to the normalised token. We can observe that this stock to market and market to stock exchange generalises a single fixed pooled context. In the simplest such coupling the market state is merely the mean embedding and it is concatenated once to each security, whereas here the pooling and the broadcast are each mediated by a learned perceptron, so the market summary and the way it re enters each security are both learned rather than fixed.

5 The omission of time mixing

The model departs deliberately from the original architecture, for a reason that follows from the structure of the data. The third mixing axis of the original StockMixer is time, along which a per stock history is mixed. The competition panel supplies a single monthly cross section of characteristics with no per stock lookback sequence attached to each formation date, so there is no time axis along which to mix within a month. The time mixing block is therefore omitted. The temporal dimension of the problem is instead handled by the rolling window and the warm started refitting, which operate at the level of the estimation protocol rather than within the architecture.

6 Evaluation and concluding remarks

On the evaluation window the model attains a Sharpe ratio of 2.280 in the volatility scaled form and 2.292 in the unscaled form. A plainer network that couples the cross section through a fixed pooled average attains 2.484 on this panel, so the learned stock mixing does not improve on the fixed coupling here, which is consistent with a broader finding on this panel that additional flexibility in the cross sectional coupling is not rewarded given the available data. We conclude that the StockMixer architecture is a coherent generalisation of the fixed pooled coupling whose extra capacity, on this panel, is absorbed by estimation error rather than converted into a higher Sharpe ratio. The model nonetheless supplies a useful component for an equal weighted ensemble with the plainer network, since its errors are not perfectly aligned with those of that model.